

med tech curriculum V1

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Med-Tech Curriculum — V1

Companion to med-tech-plan-V1.md. The plan is the master/architecture document (vision, volumes, chapter map, hour budget, conventions). **This file holds the fully-expanded, formal chapter definitions** — each chapter filled out to the §2.5 template using the §2.6 worked example as the standard, with the real topic list and per-topic theory/lab hours.

- **Status:** V1 — **all 27 chapters expanded** (first pass). Pending pedagogical review of content and hours.
 - **Source of structure:** med-tech-plan-V1.md §2.3 (Volume & Chapter map) and §2.5 (chapter template).
 - **Hour figures:** theory (T) = lecture/seminar contact hours; lab (L) = supervised hands-on hours. Per-topic hours **sum to the chapter budget** in the plan's §2.3. Figures remain **provisional** until reviewed.
 - **Progress: Complete — Volumes I–V, Chapters 1–27** (see per-volume summaries and the program grand total at the end). Core (I–IV) = 542 h; with elective Vol V = 646 h.
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Volume I — Foundations & Clinical Data Engineering (*Phase 0*)

Chapters 1–5. Volume budget: **T 58 h + L 66 h = 124 h.**

Chapter 1 — Python for Data Science (NumPy, Pandas, Matplotlib) · [Volume I]

Prerequisites: basic Python (entry skill for the program — variables, control flow, functions, lists/dicts).

Learning outcomes — the student can:

- Manipulate numerical data efficiently with **NumPy** (vectorization, broadcasting, indexing, aggregation).
- Load, clean, transform, and join tabular **clinical data** with **Pandas** (missing values, types, dates, categoricals, merges, groupby).
- Perform basic **feature engineering** to build a modeling-ready table from raw EHR-style data.
- Produce clear exploratory **visualizations** with Matplotlib (distributions, cohort comparisons, missingness).
- Work in **reproducible, well-structured notebooks** with sound data hygiene.

Topics (theory) — 12 h:

#	Topic	h
1.1	Python for data work: environment (Anaconda/conda), Jupyter, reproducibility	1
1.2	NumPy: ndarray, vectorization, broadcasting, indexing/slicing, aggregation	3
1.3	Pandas Series/DataFrame,	l: 2

#	Topic	h
	I/O (CSV/Parquet), indexing & selection	
1.4	Pandas II: cleaning — missing data, dtypes, dates, categoricals, merges/joins, groupby	3
1.5	Feature engineering for clinical/tabular data	1
1.6	Matplotlib (and pandas plotting): distributions, correlations, time series	1
1.7	Reproducible, well- structured notebooks & data hygiene	1

Labs (hands-on) — 18 h:

#	Lab	h
1a	NumPy warm-up: vitals array computations, vectorized normalization/z-scores	3
1b	Load & explore a synthetic EHR CSV in Pandas; basic profiling/EDA	3
1c	Data cleaning: missing	4

#	Lab	h
	values, dtype fixes, date parsing, deduplication	
1d	Merge multi-table EHR (patients / encounters / labs); groupby aggregations	3
1e	Feature engineering: build a per-patient modeling table	3
1f	Visualization: distributions, cohort comparison, missingness heatmap	2

Datasets/tools: synthetic EHR CSVs (or a Synthea export); NumPy, Pandas, Matplotlib, Jupyter; conda/Anaconda environment.

Assessment: notebook deliverable — a cleaned dataset + short EDA report (rubric, **60%**); quiz on NumPy/Pandas idioms (**20%**); feature-engineering task producing a modeling table (**20%**).

Key decisions taught here:

- NumPy vs. Pandas for a given operation (raw arrays vs. labeled tables).
- **Wide vs. long** data format for clinical time series.
- Handling missing clinical data — **impute vs. drop** — and its **clinical implications** (bias, leakage).
- Reproducibility choices (environment pinning, deterministic notebooks).

References: plan §13 → *Programming & data science*.

Hours: Theory **12** + Lab **18** = **30**. (*Feeds plan Lab 1 — EHR Data Wrangling.*)

Chapter 2 — Foundations of Clinical Data (EHR; code sets; privacy) · [Volume I]

Prerequisites: Ch 1 (Python/Pandas).

Learning outcomes — the student can:

- Describe **EHR structure** and its core data domains, and distinguish structured vs. unstructured data.
- Navigate and **map medical code sets** — ICD-10-CM/PCS, CPT/HCPCS, SNOMED CT, LOINC — to the right purpose.
- Explain **data privacy/regulation** (HIPAA, GDPR) and what makes clinical data identifiable.
- Apply basic **de-identification** and justify **synthetic-data** use for development.

Topics (theory) — 10 h:

#	Topic	h
2.1	The EHR: data domains (demographics, encounters, dx, procedures, meds, labs, vitals, notes); structured vs. unstructured	2
2.2	Diagnosis & procedure code sets: ICD-10-CM/PCS,	2

#	Topic	h
	CPT/HCPCS	
2.3	Clinical terminologies: SNOMED CT (concepts/hierarchies), LOINC (labs)	2
2.4	Mapping, crosswalks & value sets between terminologies	1
2.5	Privacy & regulation: HIPAA (PHI, Safe Harbor / Expert Determination), GDPR (special-category data, lawful basis)	2
2.6	De-identification, synthetic data & data-governance basics	1

Labs (hands-on) — 6 h:

#	Lab	h
2a	Explore an EHR schema; classify fields into data domains; structured vs. unstructured	2
2b	Code lookup & mapping: build a small ICD-10 ↔ SNOMED ↔ LOINC crosswalk in	2

#	Lab	h
	Pandas	
2c	De-identification: apply HIPAA Safe Harbor (18 identifiers) to a synthetic dataset; verify	2

Datasets/tools: synthetic EHR; public browsers (ICD-10, SNOMED CT browser, LOINC search); Pandas.

Assessment: quiz on code sets & privacy (**40%**); crosswalk deliverable (**30%**); de-identification lab rubric (**30%**).

Key decisions: which terminology fits which purpose; when data is/ isn't de-identified; **synthetic vs. real** data for development.

References: plan §13 → *Clinical data, code sets & privacy*.

Hours: Theory **10** + Lab **6** = **16**.

Chapter 3 — Healthcare Interoperability (HL7 v2, CDA/C-CDA, FHIR, SMART-on-FHIR) · [Volume I]

Migrated from plan §2.6 (the template worked example).

Prerequisites: Ch 1 (Python/Pandas), Ch 2 (clinical data & code sets).

Learning outcomes — the student can:

- Explain how clinical data is **exchanged** (not just stored) and why it is the backbone of medical AI.
- **Read and parse** an HL7 v2 message and a C-CDA document into structured data.

- Model clinical data as **FHIR resources** and query a FHIR server via its **REST API** from Python.
- **Bind** data to standard terminologies (ICD/SNOMED/LOINC) and describe a **SMART-on-FHIR** integration.

Topics (theory) — 12 h:

#	Topic	h
3.1	Why interoperability matters; the data backbone of medical-AI products	1
3.2	HL7 v2.x messaging: segments, trigger events (ADT/ORM/ORU), pipe-delimited format	2
3.3	HL7 CDA & C-CDA: XML clinical documents, templates, sections	2
3.4	FHIR fundamentals: resources, references, bundles, data types	3
3.5	FHIR REST API: search, read, CRUD, versioning, pagination	2
3.6	Terminology binding in FHIR (ICD/SNOMED/LOINC); profiles &	1

#	Topic	h
	conformance	
3.7	SMART-on-FHIR & app integration; OAuth scopes, security basics	1

Labs (hands-on) — 16 h:

#	Lab	h
3a	Parse an HL7 v2 message with hl7apy into structured fields	3
3b	Read a C-CDA document; extract problems / medications / allergies	3
3c	Stand up a HAPI FHIR server and load Synthea synthetic patients	3
3d	Query the FHIR REST API from Python; Patient/Observation/Condition → Pandas	4
3e	Build a patient-summary extractor over FHIR (mini SMART-on-FHIR read flow)	3

Datasets/tools: Synthea synthetic patients; HAPI FHIR (public sandbox or local); Python hl7apy, fhir.resources/fhirclient, requests, Pandas; sample C-CDA XML.

Assessment: quiz on standards (**20%**); lab rubric on 3a–3e (**50%**); a "clinical data → tidy DataFrame" deliverable from a FHIR bundle (**30%**).

Key decisions: FHIR vs. HL7 v2 for a new integration; **document (CDA) vs. resource (FHIR)** models; how much to normalize/terminology-map before feeding an AI pipeline.

References: plan §13 → *Interoperability standards*.

Hours: Theory **12** + Lab **16** = **28**.

Chapter 4 — Core Machine Learning for Medicine · [Volume I]

Prerequisites: Ch 1 (NumPy/Pandas), Ch 2 (clinical data).

Learning outcomes — the student can:

- Frame a clinical problem as **supervised or unsupervised** ML and build/train/evaluate models with **scikit-learn**.
- Apply a rigorous **workflow** (train/val/test, cross-validation) and avoid **data leakage**.
- Choose and **interpret clinically appropriate metrics** (sensitivity/specificity, PPV/NPV, ROC-AUC, PR-AUC, calibration).
- Use **unsupervised** methods (clustering, PCA) for **patient stratification**, and recognize imbalance/bias issues.

Topics (theory) — 14 h:

#	Topic	h
4.1	ML framing for medicine: supervised	1

#	Topic	h
	vs. unsupervised; clinical use cases	
4.2	The ML workflow: train/validation/test, cross-validation, data leakage	2
4.3	Supervised I: logistic regression, linear models, regularization	2
4.4	Supervised II: decision trees, random forests, gradient boosting	2
4.5	Evaluation & metrics: confusion matrix, sensitivity/specificity/PV/NPV, ROC-AUC, PR-AUC, calibration	3
4.6	Class imbalance, thresholds & clinical decision trade-offs	1
4.7	Unsupervised: clustering (k-means, hierarchical) + PCA for stratification	2
4.8	Overfitting, bias/variance & model selection	1

Labs (hands-on) — 16 h:

#	Lab	h
4a	Build a readmission classifier (logistic regression) with a proper train/test split	3
4b	Random forest / gradient boosting on the same task; compare; feature importance	3
4c	Evaluation deep-dive: ROC/PR curves, calibration, clinical threshold selection	3
4d	Cross-validation & hyperparameter tuning; leakage checks	3
4e	Unsupervised patient stratification: clustering + PCA visualization	4

Datasets/tools: synthetic structured EHR; scikit-learn, Pandas, Matplotlib.

Assessment: modeling project — build + evaluate a classifier with written interpretation (50%); evaluation/metrics lab rubric (30%); quiz (20%).

Key decisions: metric choice for the clinical question; **sensitivity vs. specificity** trade-off; complexity vs. **interpretability**; handling imbalance; avoiding leakage.

References: plan §13 → *Programming & data science*; *Clinical datasets*.

Hours: Theory **14** + Lab **16** = **30**. (*Feeds plan Lab 2 — Simple Clinical Classifier.*)

Chapter 5 — Medical NLP Basics (regex → embeddings) · [Volume I]

Prerequisites: Ch 1 (Python/Pandas).

Learning outcomes — the student can:

- Process clinical free text with **classical NLP** (tokenization, normalization, TF-IDF).
- Build **regex/rule-based extractors** for clinical concepts and detect **negation**.
- Use **word embeddings** (general and biomedical) and build a simple clinical **text classifier**.
- Articulate the **limits of classical NLP** (context, negation) that motivate transformers (bridge to Volume II).

Topics (theory) — 10 h:

#	Topic	h
5.1	Clinical characteristics & challenges (abbreviations, negation, jargon)	text: 1
5.2	Preprocessing: tokenization, normalization, stopwords, stemming/lemmatization	2

#	Topic	h
5.3	Regular expressions for clinical information extraction	2
5.4	Representations: bag-of-words, TF-IDF	1
5.5	Word embeddings: word2vec/GloVe; biomedical embeddings	2
5.6	Simple text classification, basic NER & negation detection (e.g., NegEx)	1
5.7	Limits of classical NLP → why transformers (bridge to Volume II)	1

Labs (hands-on) — 10 h:

#	Lab	h
5a	Preprocess clinical notes: tokenize, normalize, build a clean corpus	2
5b	Regex extractor: pull vitals / medications / dates from clinical text	3
5c	TF-IDF + logistic regression: classify	3

#	Lab	h
	clinical type/intent	note
5d	Explore embeddings: neighbors biomedical terms	word 2 nearest on

Datasets/tools: synthetic/de-identified clinical notes; Python re, scikit-learn, spaCy/NLTK, gensim or pretrained biomedical embeddings.

Assessment: regex extractor deliverable — rubric (**40%**); text-classification lab (**40%**); quiz (**20%**).

Key decisions: rule-based vs. statistical NLP; when regex suffices vs. needs ML; **general vs. biomedical** embeddings; handling **negation/context**.

References: plan §13 → *Transformers & the Hugging Face ecosystem* (bridge); *Programming & data science*.

Hours: Theory **10** + Lab **10** = **20**.

Volume I — completion summary

Ch	Title	T	L	Total
1	Python for Data Science	12	18	30
2	Foundations of Clinical Data	10	6	16
3	Healthcare Interoperabil	12	16	28

Ch	Title	T	L	Total
	ity			
4	Core Machine Learning for Medicine	14	16	30
5	Medical NLP Basics	10	10	20
	Volume I total	58	66	124

Matches the plan §2.3 budget. ✓

Volume II — The LLM Era (*Phase 1*)

Chapters 6–11. Volume budget: **T 56 h + L 62 h = 118 h**.

Chapter 6 — Transformer Architecture & Self-Attention · [Volume II]

Prerequisites: Ch 4 (ML), Ch 5 (NLP basics).

Learning outcomes — the student can: explain self-attention and the transformer block; describe tokenization, embeddings and positional encoding; distinguish encoder/decoder/encoder-decoder families; read a model card and reason about parameters/context length.

Topics (theory) — 12 h:

#	Topic	h
6.1	From embeddings to context; limits of classical NLP	1
6.2	Tokenization & subwords (BPE, WordPiece)	2
6.3	Self-attention & multi-head attention	3
6.4	The transformer block: FFN, residuals, layer norm, positional encoding	2
6.5	Architectures: encoder (BERT), decoder (GPT), encoder-decoder (T5)	2
6.6	Pretraining objectives (MLM, causal LM) & scaling intuition	1
6.7	Reading model cards: parameters, context length, tokens	1

Labs (hands-on) — 6 h:

#	Lab	h
6a	Tokenization exploration: tokenize clinical text, inspect subwords & token	2

#	Lab	h
	counts	
6b	Visualize attention weights on a clinical sentence	2
6c	Run inference with a small pretrained transformer; extract embeddings	2

Datasets/tools: Hugging Face transformers; a small BERT/GPT; clinical text samples. **Assessment:** architecture quiz (50%); attention-visualization writeup (30%); tokenization exercise (20%). **Key decisions:** encoder vs. decoder for a task; context length vs. cost; tokenizer choice. **References:** plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory 12 + Lab 6 = 18.

Chapter 7 — The Hugging Face Ecosystem · [Volume II]

Prerequisites: Ch 6.

Learning outcomes — the student can: navigate the Hub; use transformers pipelines and AutoModel/AutoTokenizer; load and process data with datasets; run efficient inference; manage models and licenses locally.

Topics (theory) — 8 h:

#	Topic	h
7.1	The Hub: models, datasets, spaces; model cards & licenses	1
7.2	transformers:	2

#	Topic	h
	pipelines, AutoModel/AutoTokenizer, config	
7.3	datasets: loading, map, streaming	2
7.4	Tokenizers, batching, padding/truncation	1
7.5	Inference: CPU/GPU, half precision, device management	1
7.6	The evaluate library & metrics basics	1

Labs (hands-on) — 12 h:

#	Lab	h
7a	Load a biomedical model via pipeline; run NER/classification	3
7b	Load & preprocess a medical dataset with datasets (map/tokenize)	3
7c	Batch inference over a clinical corpus; save outputs	3
7d	Build a small reproducible inference script + pinned	3

#	Lab	h
environment		

Datasets/tools: HF transformers, datasets, evaluate; PubMedBERT and similar.

Assessment: inference-pipeline deliverable (**50%**); dataset-processing lab (**30%**); quiz (**20%**). **Key decisions:** model choice/license; precision vs. memory; streaming vs. in-memory datasets.

References: plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory **8** + Lab **12** = **20**.

Chapter 8 — Prompt Engineering for Medicine · [Volume II]

Prerequisites: Ch 7.

Learning outcomes — the student can: design effective prompts (zero/few-shot, chain-of-thought); produce structured outputs; assess reliability and hallucination; apply basic safety/injection awareness in clinical apps.

Topics (theory) — 8 h:

#	Topic	h
8.1	Prompting fundamentals: zero/few-shot, roles, instructions	2
8.2	Chain-of-thought & reasoning prompts	1
8.3	Structured outputs (JSON, schemas), constrained generation	1
8.4	Retrieval-grounded prompting (intro to the	1

#	Topic	h
	RAG idea)	
8.5	Reliability, hallucination & self-consistency; when NOT to trust	2
8.6	Safety & prompt-injection awareness in clinical apps	1

Labs (hands-on) — 10 h:

#	Lab	h
8a	Prompt iteration on a clinical QA task; zero-vs. few-shot	3
8b	Chain-of-thought vs. direct on clinical reasoning; measure	2
8c	Structured extraction: prompt an LLM to output a JSON clinical summary	3
8d	Hallucination probe: design tests, log failures	2

Datasets/tools: local LLM via Ollama (MedGemma) or HF; Python. **Assessment:** prompt-portfolio deliverable (**50%**); structured-extraction lab (**30%**); quiz (**20%**). **Key decisions:** prompting vs. fine-tuning vs. RAG; how to measure reliability; guardrails. **References:** plan §13 → *Medical LLMs*. **Hours:** Theory **8** + Lab **10** = **18**.

Chapter 9 — Medical Domain Adaptation · [Volume II]

Prerequisites: Ch 7, Ch 8.

Learning outcomes — the student can: distinguish continued pretraining vs. fine-tuning vs. instruction tuning; explain how Meditron/Clinical-Camel/MedGemma were adapted; choose an adaptation strategy; reason about data requirements and licensing.

Topics (theory) — 10 h:

#	Topic	h
9.1	Why adapt general LLMs to medicine; gaps & risks	1
9.2	Continued pretraining on biomedical corpora	2
9.3	Supervised fine-tuning & instruction tuning	2
9.4	Case studies: Meditron, Clinical-Camel, MedGemma, PubMedBERT	2
9.5	Data: sources, curation, quality, licensing for medical corpora	2
9.6	Choosing an adaptation strategy vs. RAG (decision framework)	1

Labs (hands-on) — 6 h:

#	Lab	h
9a	Compare a general vs. domain-adapted model on medical prompts	2
9b	Prepare a small biomedical instruction dataset (format, clean)	2
9c	Design an adaptation plan for a chosen clinical use case	2

Datasets/tools: HF models (Meditron/MedGemma/PubMedBERT), datasets.

Assessment: adaptation-plan deliverable (**50%**); comparison lab (**30%**); quiz (**20%**).

Key decisions: pretrain vs. fine-tune vs. RAG; data licensing; base-model choice.

References: plan §13 → *Medical LLMs; Fine-tuning, alignment & benchmarks*. **Hours:** Theory **10** + Lab **6** = **16**.

Chapter 10 — Benchmarking & Hallucination · [Volume II]

Prerequisites: Ch 8, Ch 9.

Learning outcomes — the student can: evaluate LLMs on medical benchmarks (MedQA/PubMedQA/MedMCQA); interpret results; measure hallucination; build an eval harness; explain why benchmark ≠ clinical safety.

Topics (theory) — 8 h:

#	Topic	h
10.1	Benchmark	2

#	Topic	h
	landscape: MedQA (USMLE), PubMedQA, MedMCQA	
10.2	Metrics: accuracy, exact-match & their limits	1
10.3	Hallucination: definitions, causes, measurement	2
10.4	Faithfulness & grounding metrics	1
10.5	Benchmark $\neq$ clinical safety; distribution shift	2

Labs (hands-on) — 8 h:

#	Lab	h
10a	Run a model on a MedQA-style set; compute accuracy	3
10b	Build a small eval harness with the evaluate library	2
10c	Hallucination test: craft adversarial clinical prompts, score	3

Datasets/tools: HF datasets/evaluate; local LLM. **Assessment:** eval-harness deliverable (50%); hallucination report (30%); quiz (20%). **Key decisions:** which benchmark supports which claim; honest reporting of limitations. **References:** plan §13 → *Fine-tuning, alignment & benchmarks; Medical LLMs*. **Hours:** Theory 8 + Lab 8 = 16.

Chapter 11 — First Fine-Tunes (PubMedBERT, QLoRA, local MedGemma/Ollama) · [Volume II]

Prerequisites: Ch 7–10.

Learning outcomes — the student can: fine-tune a biomedical BERT for classification; apply QLoRA to fine-tune a small generative LLM on medical QA; run models locally via Ollama; evaluate a fine-tune and write a model card.

Topics (theory) — 10 h:

#	Topic	h
11.1	Fine-tuning mechanics: heads, learning rate, epochs, overfitting	2
11.2	Parameter-efficient fine-tuning: LoRA & QLoRA (quantization)	3
11.3	Datasets for fine- tuning: formatting, splits, tokenization	1
11.4	Local deployment: Ollama, GGUF, quantization levels	2
11.5	Evaluating a fine-	1

#	Topic	h
	tuned medical model; avoiding leakage	
11.6	Compute & memory budgeting on consumer GPUs	1

Labs (hands-on) — 20 h:

#	Lab	h
11a	Fine-tune PubMedBERT for clinical-question intent classification	5
11b	Prepare a medical QA dataset for generative fine-tuning	3
11c	QLoRA fine-tune a small Llama/GPT-2 on medical QA	6
11d	Run MedGemma locally via Ollama; compare base vs. fine-tuned	3
11e	Evaluate the fine-tune (quant + qual) & write a model card	3

Datasets/tools: HF transformers/peft/bitsandbytes, QLoRA, Ollama, MedGemma; GPU workstation. **Assessment:** fine-tuned model + model card (**60%**); eval report (**20%**); quiz (**20%**). **Key decisions:** full vs. PEFT fine-tuning; quantization level vs.

quality; base model/size vs. hardware. **References:** plan §13 → *Fine-tuning...*; *Medical LLMs; ...local inference*. **Hours:** Theory **10** + Lab **20** = **30**.

Volume II — completion summary

Ch	Title	T	L	Total
6	Transformer Architecture & Self- Attention	12	6	18
7	The Hugging Face Ecosystem	8	12	20
8	Prompt Engineering for Medicine	8	10	18
9	Medical Domain Adaptation	10	6	16
10	Benchmarki ng & Hallucinatio n	8	8	16
11	First Fine- Tunes	10	20	30
	Volume II total	56	62	118

Matches the plan §2.3 budget. ✓

Volume III — Autonomous Medical AI Systems (Phase 2)

Chapters 12–17. Volume budget: **T 64 h + L 136 h = 200 h.**

Chapter 12 — Retrieval-Augmented Generation (RAG) · [Volume III]

Prerequisites: Ch 7, Ch 8, Ch 11.

Learning outcomes — the student can: explain RAG architecture; build an ingest→embed→retrieve→generate pipeline; ground answers with citations; evaluate retrieval and answer faithfulness; reduce hallucination.

Topics (theory) — 12 h:

#	Topic	h
12.1	Why RAG; grounding vs. parametric knowledge	1
12.2	RAG architecture: retriever + generator; the pipeline	2
12.3	Chunking strategies for clinical documents	2
12.4	Embeddings & similarity search fundamentals	2

#	Topic	h
12.5	Prompt assembly, context windows, citation/provenance	2
12.6	RAG evaluation: retrieval metrics + answer faithfulness	2
12.7	Advanced: reranking, hybrid search, query rewriting	1

Labs (hands-on) — 18 h:

#	Lab	h
12a	Build a basic RAG over clinical guidelines (ingest→embed→retrieve→answer)	5
12b	Chunking experiments: compare strategies on retrieval quality	3
12c	Add citations/provenance to answers	3
12d	Evaluate RAG: retrieval recall + faithfulness scoring	4
12e	Add reranking / hybrid search; measure	3

#	Lab	h
	improvement	

Datasets/tools: LangChain/LlamaIndex or custom; embeddings model; Chroma; local LLM. **Assessment:** working RAG system (**60%**); eval report (**20%**); quiz (**20%**). **Key decisions:** chunk size/overlap; embedding model; RAG vs. fine-tune; citation strategy. **References:** plan §13 → *RAG, vector databases & local inference*. **Hours:** Theory **12** + Lab **18** = **30**.

Chapter 13 — Vector Databases & Knowledge Storage · [Volume III]

Prerequisites: Ch 12.

Learning outcomes — the student can: explain embeddings storage and ANN search; use Chroma/FAISS (and know Weaviate/Pinecone); design metadata & filtering; reason about where knowledge lives in hardware (RAM/NVMe) and how it scales.

Topics (theory) — 8 h:

#	Topic	h
13.1	Vector embeddings & similarity metrics (cosine, dot, L2)	1
13.2	ANN indexes: HNSW, IVF; recall vs. speed	2
13.3	Vector DBs compared: Chroma, Weaviate, Pinecone, FAISS	2
13.4	Metadata, filtering & hybrid	1

#	Topic	h
	(keyword+vector) search	
13.5	Where knowledge lives: RAM vs. NVMe; persistence; scaling	2

Labs (hands-on) — 12 h:

#	Lab	h
13a	Build & query a Chroma collection with metadata filters	3
13b	Compare FAISS index types (recall/speed)	3
13c	Persist & reload a vector store; measure memory footprint	3
13d	Hybrid search: combine BM25 + vectors	3

Datasets/tools: Chroma, FAISS, Weaviate (optional); embeddings model.

Assessment: vector-store lab (**50%**); benchmark report (**30%**); quiz (**20%**). **Key**

decisions: which vector DB; index type vs. recall/latency; RAM vs. disk; managed vs.

local. **References:** plan §13 → *RAG, vector databases & local inference*. **Hours:**

Theory **8** + Lab **12** = **20**.

Chapter 14 — Training Pipeline: Pre-training → Fine-tuning → Alignment (DPO) · [Volume III]

Prerequisites: Ch 11, Ch 12.

Learning outcomes — the student can: describe the full three-stage pipeline; reason about data/compute for pretraining; run domain fine-tuning; build preference data and apply DPO; compare RLHF vs. DPO; manage training runs with tracking.

Topics (theory) — 16 h:

#	Topic	h
14.1	The three-stage pipeline overview (pretrain→SFT→align)	1
14.2	Pretraining: data scale, objectives, tokenization, compute	2
14.3	Domain fine-tuning on medical corpora (e.g., MedFineWeb)	2
14.4	Instruction/supervised fine-tuning data design	2
14.5	Preference alignment: RLHF overview	2
14.6	Direct Preference Optimization (DPO): theory & practice	3
14.7	Safety alignment for medical outputs	2

#	Topic	h
14.8	Training checkpoints, precision, basics, experiment tracking	infra: 2 mixed distributed

Labs (hands-on) — 20 h:

#	Lab	h
14a	Continued (small) biomedical corpus pretraining on a	5
14b	Build an SFT dataset & run supervised fine-tuning	5
14c	Create a preference dataset (chosen/rejected) for a clinical behavior	3
14d	Run DPO on the SFT model	5
14e	Track experiments & compare (metrics, checks) stages safety	2

Datasets/tools: HF transformers/trl/peft, DPO; experiment tracking (W&B or MLflow); GPU. **Assessment:** aligned model across stages (**60%**); training report (**20%**); quiz (**20%**). **Key decisions:** pretrain vs. fine-tune vs. align scope; DPO vs. RLHF; data

quality vs. quantity; compute budget. **References:** plan §13 → *Fine-tuning, alignment & benchmarks; Medical LLMs; Clinical datasets*. **Hours:** Theory **16** + Lab **20** = **36**.

Chapter 15 — Evaluation, Bias, Fairness & Clinical Safety · [Volume III]

Prerequisites: Ch 10, Ch 14.

Learning outcomes — the student can: design rigorous clinical evaluation; measure subgroup bias/fairness; assess calibration and safety (red-teaming); explain the benchmark-to-clinic gap; run clinician-in-the-loop evaluation.

Topics (theory) — 12 h:

#	Topic	h
15.1	Evaluation beyond accuracy: task-appropriate metrics	2
15.2	Bias & fairness: subgroup analysis, disparate performance	2
15.3	Calibration & uncertainty in clinical models	2
15.4	Safety: harmful outputs, refusal, red-teaming	2
15.5	Benchmark vs. clinical validity; prospective vs. retrospective	2
15.6	Human evaluation &	2

#	Topic	h
	clinician-in-the-loop protocols	

Labs (hands-on) — 10 h:

#	Lab	h
15a	Subgroup fairness analysis on a clinical model	3
15b	Calibration curves & reliability diagrams	2
15c	Red-team a medical LLM; catalog failure modes	3
15d	Design a human-eval rubric & run a mini study	2

Datasets/tools: evaluate, fairness libraries; local LLM. **Assessment:** safety/fairness eval report (60%); red-team catalog (20%); quiz (20%). **Key decisions:** which subgroups & metrics; acceptable risk thresholds; when a model is **not** safe to deploy. **References:** plan §13 → *Medical LLMs; Fine-tuning, alignment & benchmarks*. **Hours:** Theory 12 + Lab 10 = 22.

Chapter 16 — Deployment, FHIR Integration & Compliance · [Volume III]

Prerequisites: Ch 12, Ch 15, Ch 3.

Learning outcomes — the student can: deploy models locally (Ollama/Open WebUI); build an inference API; integrate via FHIR (read/write, SMART-on-FHIR); monitor performance/drift; navigate the regulatory/compliance landscape.

Topics (theory) — 10 h:

#	Topic	h
16.1	Deployment options: local serving, Ollama, Open WebUI, APIs	2
16.2	Inference serving: latency, batching, quantization, scaling	2
16.3	FHIR integration & SMART-on-FHIR app model (read/write)	2
16.4	Monitoring, logging, drift & feedback loops	2
16.5	Regulatory & compliance landscape (SaMD, FDA/CE, HIPAA/GDPR at deployment)	2

Labs (hands-on) — 16 h:

#	Lab	h
16a	Deploy a model with Ollama + Open WebUI	3
16b	Wrap the model in a REST inference API	4

#	Lab	h
	(FastAPI)	
16c	FHIR integration: read patient context, write results back	5
16d	Add monitoring/logging & a drift check	4

Datasets/tools: Ollama, Open WebUI, FastAPI, HAPI FHIR/Synthea; monitoring tools.

Assessment: deployed FHIR-integrated service (**60%**); monitoring setup (**20%**); quiz (**20%**). **Key decisions:** local vs. cloud; API design; regulatory pathway; PHI handling at inference. **References:** plan §13 → *...local inference*; *Interoperability standards*; *Clinical data, code sets & privacy*. **Hours:** Theory **10** + Lab **16** = **26**.

Chapter 17 — Capstone Project · [Volume III]

Prerequisites: all of Volumes I–III.

Learning outcomes — the student can: independently design, build, evaluate, and deploy an end-to-end medical-AI system for a real clinical problem; document decisions and safety; present with evidence to clinical stakeholders.

Topics (theory) — 6 h:

#	Topic	h
17.1	Scoping a clinical AI problem; requirements & success metrics	2
17.2	Project planning, data & licensing, risk/safety	2

#	Topic	h
	plan	
17.3	Documentation, model cards & presenting to clinical stakeholders	2

Project (hands-on) — 60 h:

#	Milestone	h
17a	Problem definition & data/knowledge sourcing	8
17b	Build knowledge base / RAG pipeline	12
17c	Train/adapt the model (QLoRA/DPO as needed)	14
17d	Evaluation & safety/fairness assessment	10
17e	Deploy locally with FHIR-style I/O (inference server)	10
17f	Documentation, model card & final presentation	6

Example project: "Clinical Trial Matching Assistant" (plan §5.2). **Datasets/tools:** the full stack from Volumes I–III. **Assessment:** end-to-end system (**50%**); evaluation & safety documentation (**25%**); presentation/defense (**25%**). **Key decisions:** the full stack

of trade-offs — RAG vs. fine-tune, hardware, scope vs. safety. **References:** plan §5.2; §13 (all). **Hours:** Theory **6** + Project **60** = **66**.

Volume III — completion summary

Ch	Title	T	L	Total
12	Retrieval-Augmented Generation (RAG)	12	18	30
13	Vector Databases & Knowledge Storage	8	12	20
14	Training Pipeline (pretrain→fine-tune→DP O)	16	20	36
15	Evaluation, Bias, Fairness & Clinical Safety	12	10	22
16	Deployment, FHIR Integration & Compliance	10	16	26
17	Capstone	6	60	66

Ch	Title	T	L	Total
	Project			
	Volume III	64	136	200
	total			

Matches the plan §2.3 budget. ✓

Volume IV — From Books to LLM (*Strategic §6*)

Chapters 18–22. Volume budget: **T 38 h + L 62 h = 100 h**.

Chapter 18 — Ingestion, OCR & Document Understanding · [Volume IV]

Prerequisites: Ch 5, Ch 12.

Learning outcomes — the student can: parse born-digital and scanned medical books; run OCR handling medical vocabulary; extract layout/structure (sections, tables, references); build a clean, provenance-tagged text corpus.

Topics (theory) — 8 h:

#	Topic	h
18.1	Document types & challenges: PDFs, scans, layout, columns	2
18.2	OCR fundamentals & medical-text pitfalls (terms, dosages,	2

#	Topic	h
	tables)	
18.3	Layout analysis & structure extraction (sections, headings, refs)	2
18.4	Tables, equations & footnotes; reading order	1
18.5	Provenance & metadata capture during ingestion	1

Labs (hands-on) — 14 h:

#	Lab	h
18a	Extract text from a born-digital medical PDF (PyMuPDF)	3
18b	OCR a scanned page; measure & clean errors (Tesseract / doc model)	4
18c	Layout-aware parsing (unstructured/Nougat/Marker) into sections	4
18d	Build a provenance-tagged text corpus (page/section metadata)	3

Datasets/tools: licensed/open medical texts; PyMuPDF, Tesseract, unstructured/Nougat/Marker. **Assessment:** ingestion pipeline + corpus (60%); OCR-quality report (20%); quiz (20%). **Key decisions:** born-digital vs. scanned handling; OCR engine; how much structure to preserve; **licensing (owned/licensed sources only)**. **References:** plan §6; §13. **Hours:** Theory 8 + Lab 14 = 22.

Chapter 19 — Figures & Multimodal Understanding · [Volume IV]

Prerequisites: Ch 18, Ch 7.

Learning outcomes — the student can: detect/extract figures; generate & curate captions with multimodal models; link images to text; make images retrievable; describe the SOTA and limits of medical image understanding.

Topics (theory) — 8 h:

#	Topic	h
19.1	Medical figures: radiology, histology, ECG, diagrams, charts	1
19.2	Figure detection & extraction from documents	1
19.3	Multimodal models (captioning, VQA): LLaVA-Med, MedGemma-multimodal	2
19.4	Image embeddings & multimodal retrieval	2

#	Topic	h
19.5	Linking figures to surrounding text & captions	1
19.6	SOTA & limits: reliability of medical image understanding	1

Labs (hands-on) — 14 h:

#	Lab	h
19a	Extract figures + captions from a medical document	3
19b	Generate captions with a multimodal model; compare to originals	4
19c	Multimodal retrieval: query text → return relevant figures	4
19d	Link figures to text chunks; evaluate linkage quality	3

Datasets/tools: multimodal HF models (LLaVA-Med/MedGemma), CLIP-style embeddings; vector DB. **Assessment:** multimodal ingest+retrieval (**60%**); caption-quality report (**20%**); quiz (**20%**). **Key decisions:** which multimodal model; how much to trust captions; image vs. caption embeddings. **References:** plan §6; §13 → *Medical LLMs*. **Hours:** Theory **8** + Lab **14** = **22**.

Chapter 20 — Structuring, Provenance & Building the Corpus · [Volume IV]

Prerequisites: Ch 18, Ch 19.

Learning outcomes — the student can: semantically chunk book content; attach citations/provenance; build a versioned, updatable knowledge base; manage licensing/governance of sources.

Topics (theory) — 6 h:

#	Topic	h
20.1	Semantic chunking for books/atlasses	1
20.2	Provenance model: book/edition/page/figure citations	2
20.3	Versioning & updatability of medical knowledge	1
20.4	Licensing, ownership & governance of source content	2

Labs (hands-on) — 12 h:

#	Lab	h
20a	Build a semantic chunker with citation metadata	4
20b	Assemble a versioned knowledge base (text	4

#	Lab	h
	+ figures)	
20c	Implement citation- traceable retrieval (answer → source page)	4

Datasets/tools: vector DB; chunking libraries; a metadata store. **Assessment:** cited knowledge base (**60%**); provenance audit (**20%**); quiz (**20%**). **Key decisions:** chunk granularity; provenance schema; licensing compliance; update strategy. **References:** plan §6; §13 → RAG.... **Hours:** Theory **6** + Lab **12** = **18**.

Chapter 21 — RAG-vs-Fine-tune Paths & Synthetic Data · [Volume IV]

Prerequisites: Ch 11, Ch 12, Ch 20.

Learning outcomes — the student can: decide between RAG and fine-tuning for book knowledge; generate quality-controlled synthetic Q&A/instruction data from sources; build image-text pairs; compare accuracy/hallucination of both paths.

Topics (theory) — 8 h:

#	Topic	h
21.1	Decision framework: RAG vs. fine-tune vs. both for book knowledge	2
21.2	Synthetic Q&A generation from source text (quality	2

#	Topic	h
	control)	
21.3	Image-text pair construction for multimodal training	1
21.4	Risks: hallucinated synthetic data, contamination, licensing	2
21.5	Comparing paths: accuracy, faithfulness, cost	1

Labs (hands-on) — 16 h:

#	Lab	h
21a	Generate synthetic Q&A from a book chapter; QC filter	4
21b	Fine-tune a small model on the synthetic set (QLoRA)	5
21c	Build a RAG baseline over the same content	3
21d	Head-to-head: accuracy & hallucination, RAG vs. fine-tune	4

Datasets/tools: local LLM; HF peft/QLoRA; vector DB; eval harness. **Assessment:** comparative study (**60%**); synthetic-data QC report (**20%**); quiz (**20%**). **Key decisions:**

RAG vs. fine-tune vs. both; synthetic-data quality gates; licensing of derived data.
References: plan §6; §13 → *Fine-tuning...*; *RAG...* **Hours:** Theory 8 + Lab 16 = 24.

Chapter 22 — State of the Art & Evaluation (text/image/video) · [Volume IV]

Prerequisites: Ch 10, Ch 15, Ch 19.

Learning outcomes — the student can: survey current medical LLM/multimodal/video SOTA honestly; design evaluation for book-derived models; assess source-fidelity and citation correctness; articulate maturity gaps.

Topics (theory) — 8 h:

#	Topic	h
22.1	Text medical LLMs today: Med-PaLM 2, Meditron, MedGemma; capabilities	2
22.2	Multimodal (image+text): Med- PaLM M, LLaVA-Med; the reliability gap	2
22.3	Video understanding: surgical/echo/endosco- py — early state	2
22.4	Evaluating book- derived models: source-fidelity & citation correctness	2

Labs (hands-on) — 6 h:

#	Lab	h
22a	Fidelity eval: does the model's answer match the source book?	3
22b	Citation-correctness scoring on RAG answers	3

Datasets/tools: eval harness; the RAG/fine-tuned models built in Volume IV. **Assessment:** SOTA briefing + fidelity eval (**60%**); citation-accuracy report (**20%**); quiz (**20%**). **Key decisions:** honest capability claims; which modality is production-ready; how to report limits. **References:** plan §6.3; §13 → *Medical LLMs*. **Hours:** Theory **8** + Lab **6** = **14**.

Volume IV — completion summary

Ch	Title	T	L	Total
18	Ingestion, OCR & Document Understanding	8	14	22
19	Figures & Multimodal Understanding	8	14	22
20	Structuring, Provenance & Building	6	12	18

Ch	Title	T	L	Total
	the Corpus			
21	RAG-vs- Fine-tune Paths & Synthetic Data	8	16	24
22	State of the Art & Evaluation (text/image/ video)	8	6	14
	Volume IV total	38	62	100

Matches the plan §2.3 budget. ✓

Volume V — AI for Medical Robots (*Strategic §7 — elective/frontier*)

Chapters 23–27. Volume budget: **T 44 h + L 60 h = 104 h.**

Chapter 23 — Robotics Fundamentals & Simulation (ROS 2, Isaac Sim) · [Volume V]

Prerequisites: Ch 4; general programming maturity.

Learning outcomes — the student can: explain robot kinematics/control basics; use ROS 2 concepts (nodes, topics, services, actions); run Gazebo/Isaac Sim; command a simulated arm.

Topics (theory) — 10 h:

#	Topic	h
23.1	Robotics basics: kinematics, DoF, coordinate frames	2
23.2	Control loops & real-time constraints	2
23.3	ROS 2 architecture: nodes, topics, services, actions	3
23.4	Simulation: Gazebo & NVIDIA Isaac Sim	2
23.5	Medical robotics landscape (surgical, assistive, lab automation)	1

Labs (hands-on) — 16 h:

#	Lab	h
23a	Set up ROS 2 + a simulator; run a demo	4
23b	Publish/subscribe: command a simulated arm via topics	4
23c	Bring up a simulated robot arm in Isaac	4

#	Lab	h
	Sim; basic motion	
23d	Scripted pick task in simulation	4

Datasets/tools: ROS 2, Gazebo, Isaac Sim; GPU workstation. **Assessment:** simulation task deliverable (**60%**); ROS 2 quiz (**20%**); writeup (**20%**). **Key decisions:** simulator choice; **sim vs. hardware**; real-time constraints. **References:** plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **16** = **26**.

Chapter 24 — Perception for Clinical/Surgical Scenes · [Volume V]

Prerequisites: Ch 23; Ch 6–7 (vision-capable models).

Learning outcomes — the student can: apply computer vision to surgical/clinical scenes; segment/track instruments & anatomy; estimate depth/pose; integrate perception into a robot loop (in simulation).

Topics (theory) — 8 h:

#	Topic	h
24.1	CV for surgical scenes: occlusion, lighting, deformation	2
24.2	Segmentation & instrument tracking	2
24.3	Depth & pose estimation	2
24.4	Integrating perception into control	2

#	Topic	h
(perception→action)		

Labs (hands-on) — 14 h:

#	Lab	h
24a	Segment instruments/targets in a simulated surgical scene	4
24b	Track an instrument across frames	4
24c	Depth/pose estimation on sim data	3
24d	Close a simple perception→action loop in sim	3

Datasets/tools: OpenCV, segmentation models; the simulator. **Assessment:** perception pipeline (**60%**); tracking report (**20%**); quiz (**20%**). **Key decisions:** model choice vs. latency; sim-to-real gap; **safety of perception errors**. **References:** plan §7; §13 → *Robotics & embodied AI; Medical LLMs* (vision). **Hours:** Theory **8** + Lab **14** = **22**.

Chapter 25 — Learning for Control (Imitation/RL, sim-to-real) · [Volume V]

Prerequisites: Ch 23, Ch 24.

Learning outcomes — the student can: explain imitation learning & RL for control; train a policy in simulation; reason about sim-to-real transfer and its risks; apply safety constraints/shields.

Topics (theory) — 10 h:

#	Topic	h
25.1	Control paradigms: classical vs. learned	1
25.2	Imitation learning (behavioral cloning)	2
25.3	Reinforcement learning basics for control	3
25.4	Sim-to-real transfer: domain randomization, gaps	2
25.5	Safety constraints & constrained policies	2

Labs (hands-on) — 14 h:

#	Lab	h
25a	Behavioral cloning on a sim manipulation task	4
25b	Train an RL policy in sim (reach/grasp)	5
25c	Domain-randomization experiment	3
25d	Add a safety constraint / shield to the policy	2

Datasets/tools: RL libraries (Stable-Baselines3 / Isaac Gym); the simulator.
Assessment: trained policy (**60%**); sim-to-real analysis (**20%**); quiz (**20%**). **Key decisions:** IL vs. RL; reward design; safety shielding; when sim is insufficient.
References: plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **14** = **24**.

Chapter 26 — LLMs/VLA as Planners & Human-in-the-Loop Safety · [Volume V]

Prerequisites: Ch 11, Ch 25.

Learning outcomes — the student can: use LLMs/VLA models as high-level planners; translate NL tasks into verified action sequences; design human-in-the-loop approval; describe VLA SOTA and its limits in medicine.

Topics (theory) — 8 h:

#	Topic	h
26.1	LLMs as planners/reasoners for robots	2
26.2	Vision-Language-Action (VLA) models: concepts & SOTA (e.g., RT-2)	2
26.3	Grounding language to actions & verification	2
26.4	Human-in-the-loop: approval gates, oversight, fail-safes	2

Labs (hands-on) — 12 h:

#	Lab	h
26a	LLM task planner: NL instruction → action plan in sim	4
26b	Add verification/validation of generated plans	4
26c	Human-approval gate before execution; log & audit	4

Datasets/tools: local LLM; the simulator; a planning framework. **Assessment:** LLM-planner-in-sim with approval gate (**60%**); safety analysis (**20%**); quiz (**20%**). **Key decisions:** autonomy level; verification strategy; **when human approval is mandatory**. **References:** plan §7; §13 → *Robotics & embodied AI; Medical LLMs*. **Hours:** Theory 8 + Lab 12 = 20.

Chapter 27 — Regulation, Ethics & Liability · [Volume V]

Prerequisites: Ch 15, Ch 26.

Learning outcomes — the student can: navigate device regulation for autonomous medical systems; analyze liability/accountability; apply an ethics framework; draft a safety/validation case.

Topics (theory) — 8 h:

#	Topic	h
27.1	Medical device regulation: SaMD,	2

#	Topic	h
	FDA/CE for AI & robotics	
27.2	Autonomy levels & regulatory implications	1
27.3	Liability & accountability for autonomous action	2
27.4	Ethics: consent, equity, transparency, human oversight	2
27.5	Building a safety/validation case	1

Labs (hands-on) — 4 h:

#	Lab	h
27a	Draft regulatory/safety case for a proposed medical-robot feature	a 2
27b	Ethics & liability analysis of an autonomy scenario	2

Datasets/tools: regulatory guidance documents; case templates. **Assessment:** safety/regulatory case (**60%**); ethics analysis (**20%**); quiz (**20%**). **Key decisions:** autonomy vs. oversight; regulatory pathway; acceptable risk & accountability. **References:** plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **8** + Lab **4** = **12**.

Volume V — completion summary

Ch	Title	T	L	Total
23	Robotics Fundamentals & Simulation	10	16	26
24	Perception for Clinical/Surgical Scenes	8	14	22
25	Learning for Control (Imitation/RL , sim-to-real)	10	14	24
26	LLMs/VLA as Planners & HITL Safety	8	12	20
27	Regulation, Ethics & Liability	8	4	12
	Volume V total	44	60	104

Matches the plan §2.3 budget. ✓

Program grand total

Volume	Title	T	L	Total
I	Foundations & Clinical Data Engineering	58	66	124
II	The LLM Era	56	62	118
III	Autonomous Medical AI Systems	64	136	200
IV	From Books to LLM	38	62	100
	Core (I–IV)	216	326	542
V	AI for Medical Robots (<i>elective</i>)	44	60	104
	With elective (I– V)	260	386	646

All 27 chapters expanded. Hours match the plan §2.3 budget. Figures remain provisional pending pedagogical review.